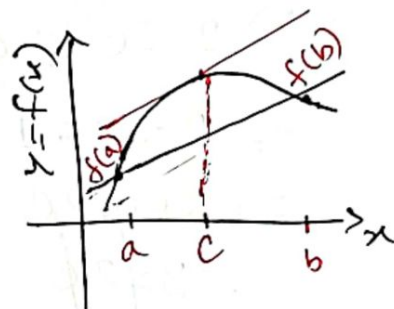


Mean Value Theorem (MVT)

Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . Then there is at least one point c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Ex 09

Verify the mean value theorem for the function $f(x) = 3 + 2x - x^2$ in the interval $(0, 1)$.

Solution: continuity: The function $f(x) = 3 + 2x - x^2$ is a polynomial. polynomials are continuous everywhere.

Therefore, $f(x)$ is continuous on $[0, 1]$.

Differentiability:

$$f(x) = 3 + 2x - x^2$$

$$f'(x) = 2 - 2x$$

$f(x)$ is differentiable on $(0, 1)$.

The hypotheses of the mean value theorem are satisfied.

Let,

$c \in (0, 1)$ such that,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow 2 - 2c = \frac{f(1) - f(0)}{1 - 0}$$

$$\Rightarrow 2 - 2c = 4 - 3$$

$$\Rightarrow 2 - 2c = 1$$

$$\Rightarrow \cancel{2+1} - 2c$$

$$\Rightarrow 3 = 2c$$

$$\Rightarrow 2c = 3$$

$$\therefore c = \frac{3}{2}$$

$$\Rightarrow 2 - 1 = 2c$$

$$\Rightarrow 1 = 2c$$

$$\therefore c = \frac{1}{2}$$

$$\therefore c = \frac{1}{2} \in (0, 1)$$

$\therefore$ (verified)

$$\text{Hence, } f(x) = 3 + 2x - x^2$$

$$f'(x) = 2 - 2x$$

$$f'(c) = 2 - 2c$$

$$a = 0$$

$$f(a) = f(0) = 3$$

$$b = 1$$

$$\begin{aligned} f(1) &= 3 + 2(1) - (1)^2 \\ &= 3 + 2 - 1 \\ &= 4 \end{aligned}$$

Ex 10

Verify the hypotheses of Mean value theorem for the function $f(x) = \sqrt{4x-3}$ over the interval $[1, 3]$, and find all values of c in the interval that satisfy the conclusion of the theorem.

Solution:

Continuity

$f(x) = \sqrt{4x-3}$ is continuous if $\sqrt{4x-3} \geq 0$

$$\therefore \sqrt{4x-3} \geq 0$$

$$\Rightarrow 4x-3 \geq 0$$

$$\therefore x \geq \frac{3}{4}$$

Hence, $x \geq \frac{3}{4}$ is the condition for the continuity of $f(x)$.

$\therefore f(x)$ is continuous on $[1, 3]$

Differentiability:

$$f(x) = \sqrt{4x-3}$$

$$f'(x) = \frac{1}{2\sqrt{4x-3}} \quad (4)$$

$$= \frac{2}{\sqrt{4x-3}}$$

$$\left| \begin{array}{l} \text{Hence,} \\ \sqrt{4x-3} > 0 \\ \Rightarrow 4x-3 > 0 \\ \therefore x > \frac{3}{4} \end{array} \right.$$

$\therefore f(x)$ is differentiable on $(1, 3)$

Let,

$$c \in (1, 3)$$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = \frac{f(3) - f(1)}{3-1}$$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = \frac{3-1}{3-1}$$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = 1$$

$$\Rightarrow 2 = \sqrt{4c-3}$$

$$\Rightarrow 4 = 4c-3$$

$$\Rightarrow 7 = 4c$$

$$c = \frac{7}{4}$$

$$\therefore c = \frac{7}{4} \in (1, 3)$$

(verified)

H.W

1) Practice sheet

2) Exercise $\rightarrow (5-8) \Rightarrow$ Anton Book (Chapter 9.8)

Hence,

$$f'(x) = \frac{2}{\sqrt{4x-3}}$$

$$f'(c) = \frac{2}{\sqrt{4c-3}}$$

$$f(x) = \sqrt{4x-3}$$

$$a = 1$$

$$b = 3$$

$$f(1) = \sqrt{4(1)-3}$$

$$= \sqrt{4-3}$$

$$= 1$$

$$f(3) = \sqrt{4(3)-3}$$

$$= \sqrt{12-3}$$

$$= \sqrt{9}$$

$$= 3$$

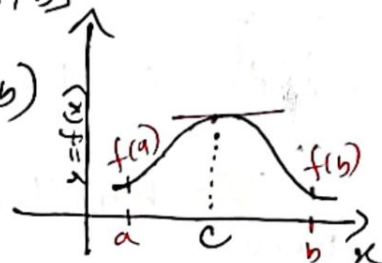
Rolle's Theorem

If a function $y=f(x)$ is

i) continuous on closed interval $[a, b]$

ii) differentiable on open interval (a, b)

iii) $f(a) = f(b)$



Then there is at least a value c in the interval (a, b) such that $f'(c) = 0$.

Ex 11

Verify the hypothesis of Rolle's theorem for the following function $f(x) = x^2 + 5x - 6$ in the interval $(-6, 1)$

continuity:

$f(x) = x^2 + 5x - 6$ is a polynomial, polynomials are continuous everywhere.

Therefore, $f(x)$ is continuous on $[-6, 1]$

Differentiability:

$$f(x) = x^2 + 5x - 6$$

$$f'(x) = 2x + 5$$

$$a = -6, b = 1$$

$$\begin{aligned} f(a) &= (-6)^2 + 5(-6) - 6 \\ &= 36 - 30 - 6 \\ &= 0 \end{aligned}$$

$$\begin{aligned} b &= 1, \\ f(b) &= (1)^2 + 5(1) - 6 \\ &= 0 \end{aligned}$$

$f(x)$ is differentiable on $(-6, 1)$. $\therefore f(a) = f(b)$

The hypotheses of Rolle's theorem are satisfied.
Now,

$$f'(x) = 2x + 5$$

$$\text{Let, } c \in (-6, 1)$$

$$f'(c) = 2c + 5$$

$$\text{Let, } f'(c) = 0$$

$$\Rightarrow 2c + 5 = 0$$

$$\therefore c = -\frac{5}{2}$$

$$\therefore c = -\frac{5}{2} \in (-6, 1)$$

(verified)

Ex 12

Verify the hypothesis of Rolle's Theorem for the following function $f(x) = 2x^3 + x^2 - 4x - 2$ in the interval $(-\sqrt{2}, \sqrt{2})$

Solution:

Continuity:

$$f(x) = 2x^3 + x^2 - 4x - 2 \text{ polynomial}$$

$\therefore f(x)$ is continuous on $[-\sqrt{2}, \sqrt{2}]$

differentiability:

$$f(x) = 2x^3 + x^2 - 4x - 2$$

$$f'(x) = 6x^2 + 2x - 4$$

$f(x)$ is differentiable on $(-\sqrt{2}, \sqrt{2})$

Now, $a = -\sqrt{2}$, $b = \sqrt{2}$

$$\begin{aligned} f(a) &= 2(-\sqrt{2})^3 + (-\sqrt{2})^2 - 4(-\sqrt{2}) - 2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} f(b) &= 2(\sqrt{2})^3 + (\sqrt{2})^2 - 4(\sqrt{2}) - 2 \\ &= 0 \end{aligned}$$

$$\therefore f(a) = f(b)$$

The hypothesis of Rolle's theorem is satisfied.

Hence,

$$f(x) = 2x^3 + x^2 - 4x - 2$$

$$f'(x) = 6x^2 + 2x - 4$$

Let,

$$c \in (-\sqrt{2}, \sqrt{2})$$

$$f'(c) = 6c^2 + 2c - 4$$

Let,

$$f'(c) = 0$$

$$\Rightarrow 6c^2 + 2c - 4 = 0$$

$$\Rightarrow 3c^2 + c - 2 = 0$$

$$\therefore c = \frac{-1 \pm \sqrt{(1)^2 - 4 \cdot 3 \cdot (-2)}}{2(3)}$$

$$= \frac{-1 \pm \sqrt{1+24}}{6} = \frac{-1 \pm \sqrt{25}}{6}$$

$$= \frac{-1 \pm \sqrt{5}}{6} = \frac{-1 \pm 5}{6}$$

$$= \frac{-1+5}{6}$$

$$= \frac{-1-5}{6}$$

$$= \frac{4}{6}, -\frac{6}{6}$$

$$= \frac{2}{3}, -1$$

$$c = \frac{2}{3} \in (-\sqrt{2}, \sqrt{2})$$

and $c = -1 \in (-\sqrt{2}, \sqrt{2})$

(verified)

H.w

1) Practice sheet

2) Exercise $\Rightarrow (1-4) \rightarrow$ Anton Book (chapter 4.8)